

## EDUCATION

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- **McGill University** Montreal, Canada  
*Bachelor Mathematics and Computer Science; CGPA: 3.7/4.0* 2023 – 2027
  - **Relevant Classes:** PDE's, Statistics, Algebra 2 (Linear Algebra), Analysis 2, Advanced Calculus, Non Linear Dynamics and Chaos, Statistical Computing, Introduction to Data Science, Stochastic Processes.
- **French Baccalaureate** Colombes, France  
*Highest Honours (18/20) – Mathematics and Economics*

## EXPERIENCE

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- **McGill University** Montreal, QC  
*Research Assistant – Probability & Random Trees* Winter 2026
  - **Random Tree Height Bounding:** Investigated the height of uniformly random rooted trees under the supervision of Prof. Masoud Asgharian, demonstrating that sub-binary trees stochastically bound the heights of trees with general degree sequences.
  - **Bijective Combinatorics:** Applied a bijection introduced by Foata and Fuchs to map tree structures to compressed degree sequences, enabling the probabilistic analysis of tree heights.
  - **Distributional Convergence:** Proved that the scaled height of the least labeled leaf has subgaussian tails and converges in distribution to a standard Rayleigh random variable.
  - **Stochastic Dominance:** Established that the height of leaves stochastically dominates the height of internal vertices within full binary trees.
- **April GROUP** Lyon, France  
*Actuarial Intern — Health & Pension Plans* May 2025 - July 2025
  - **Health Reform Simulation Tool:** Designed and implemented a simulation tool to model the impact of upcoming French health reforms on contribution rates, enabling assessment of financial and regulatory scenarios.
  - **Data Visualization & Reporting:** Developed interactive dashboards (Bokeh) and automated reporting pipelines to communicate key insights to senior actuaries and support decision-making.
  - **GLM Reserving Model:** Built and validated a GLM reserving model (Gamma-log) for incremental claims using large datasets; performed feature engineering, diagnostics, and generated bootstrapped reserve distributions for risk assessment.
  - **Pricing Analysis:** Performed data cleaning, exploratory data analysis, and trend modeling to support pricing and experience studies, improving actuarial assumptions.
- **BNP Paribas** France  
*Wealth Manager* July, 2024 - August, 2024
  - **Asset Allocation & Analysis:** Optimized high-net-worth portfolios by evaluating multi-asset financial products (equities, fixed income, structured products) and conducting quantitative assessments of market trends.
  - **Client Reporting:** Prepared detailed performance reviews and attribution analyses, translating complex portfolio returns, risk metrics, and market impacts into clear insights for high-net-worth clients.

## PROJECTS

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- **Power Grid Virtual Bidding Statistical Arbitrage (Personal Project):** Developed a quantitative trading pipeline exploiting the Day-Ahead (SDAC) to Intraday (ID3) spread on the French and German EPEX SPOT markets. Engineered a two-stage hybrid architecture combining a gradient-boosted classifier (LightGBM) to predict spread direction using 15-minute fundamental grid features (ENTSO-E API, renewable forecast errors) and a rolling volatility model for dynamic position sizing. Built a vectorized backtester in Python (pandas) to simulate PnL and assess risk-adjusted returns (Sharpe Ratio, Maximum Drawdown) net of execution costs.
- **Predicting VIX Index Movements via Analyst Sentiment Analysis (Personal Project):** Developed a predictive model for the Volatility Index (VIX) using textual data from institutional macro research reports (2017–2024). Applied Latent Dirichlet Allocation (LDA) for topic modeling, identifying inflation as the dominant market driver. Engineered sentiment features (risk, uncertainty, tone) using the Loughran–McDonald dictionary and built a non-linear regression model with interaction terms, achieving an accuracy of 85% of VIX variation.